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Problem

- Vision delta-tuning saves parameters but had not consistently surpassed full fine-tuning on dense prediction.
- NLP-style linear adapters ignore 2D visual structure and receive biased features from frozen layers.
- The question: can a small trainable module preserve pretrained knowledge and raise the performance ceiling?

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Mathematics

COMPACT FORM

- $x_{norm} = s_1 \text{LN}(x_0) + s_2 x_0$
- $x = x_0 + U_l \text{GeLU}[f_{pw}(f_{dw}(D_l(x_{norm})))]$
- Per-module parameters: $(2n+3)m + n^2 + 84n + 2$; only adapters and task heads are updated.

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Core idea

VISUAL-NATIVE ADAPTATION

- Add Mona after attention and MLP in every Swin block while freezing pretrained weights.
- Scaled LayerNorm corrects the distribution and balances normalized versus original features.
- Parallel 3 x 3, 5 x 5 and 7 x 7 depthwise filters capture multiple visual receptive fields.

MONA TUNING

5% > 100%: Breaking the Performance Shackles of Full Fine-Tuning

Multi-cognitive Visual Adapter for Recognition Tasks | 2024

Freeze the pretrained vision backbone and train a visual-native adapter that normalizes incoming features and processes them at multiple convolutional scales.

visual PEFT

multi-scale

adapter

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Evidence

5% BEATS 100%

- COCO / Swin-B: 4.67% trainable backbone parameters yield 53.40 APbox vs 52.40 for full tuning.
- Pascal VOC: 87.30 vs 83.70 APbox; ADE20K: 51.36 vs 51.18 mIoU.
- Mona also leads on DOTA/STAR, classification and multiple backbone sizes, with faster convergence.

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Mona block

INTERNAL PIPELINE

- Input optimization -> down projection -> three depthwise convolutions -> average -> 1 x 1 convolution.
- GeLU and up projection restore the channel dimension; four skip connections strengthen adaptation.
- A 64-dimensional bottleneck gives the best accuracy/parameter trade-off in ablations.

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Meaning & limits

TAKEAWAY

- For vision transfer, module design matters more than simply updating more parameters.
- Mona retains large-model knowledge, saves storage and improves as the backbone scales.
- Adapters add a small inference cost; future work could combine Mona with re-parameterization for zero overhead.